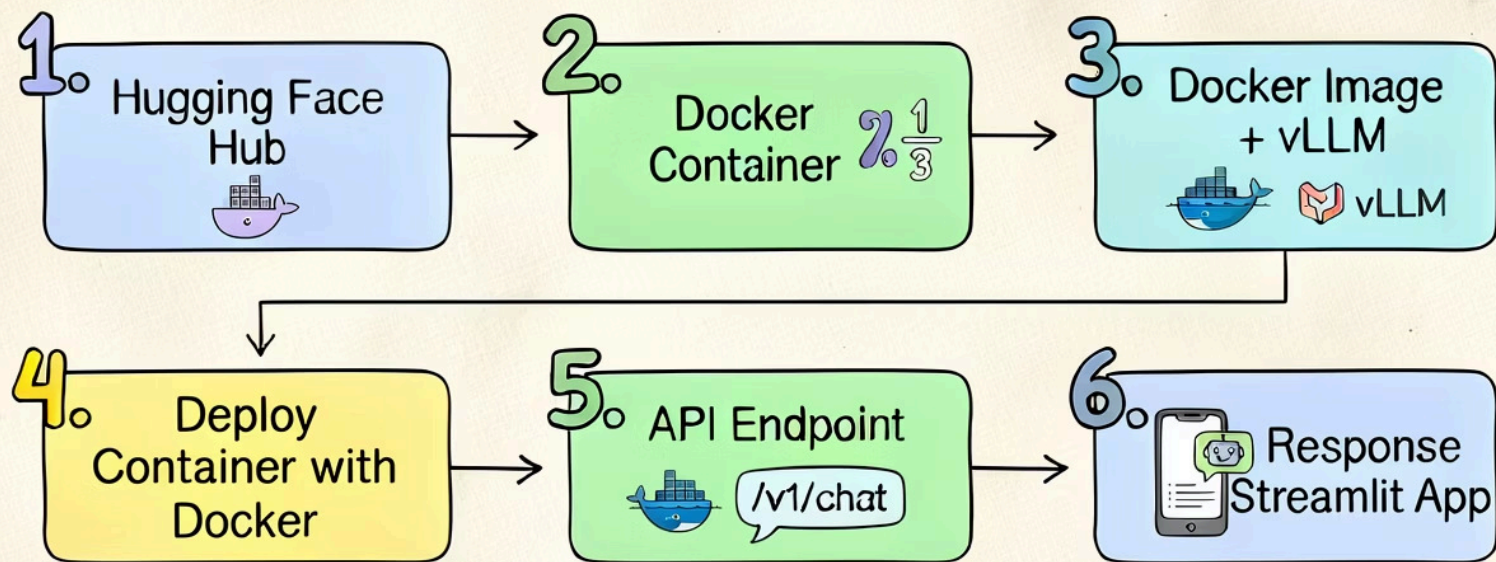


# Deploy an LLM on Kubernetes with vLLM

## Model Deployment Pipeline

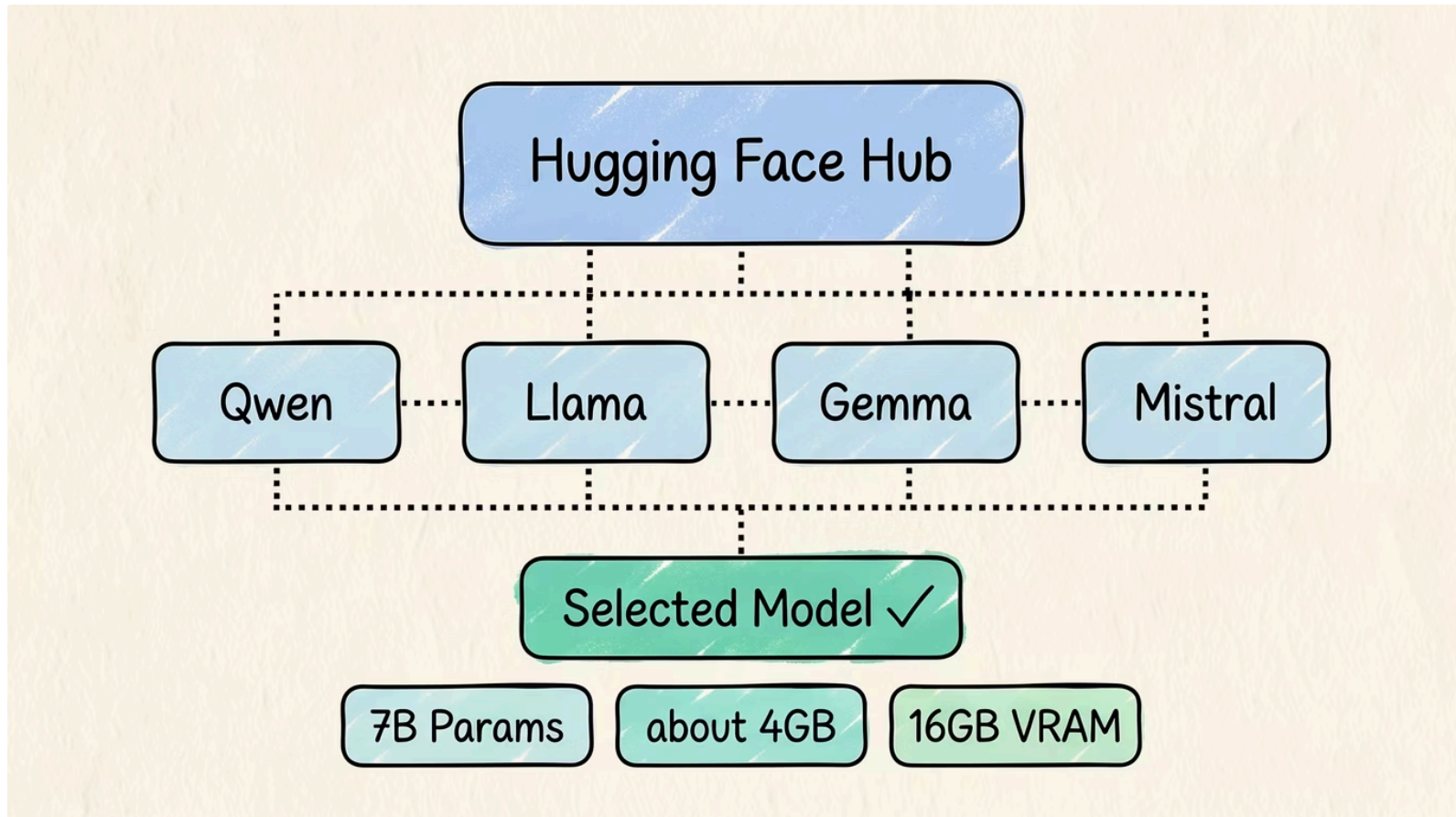


## 🤗 Step 1

### Select the Model

Pick the right open-source LLM before anything else.

Start small. A 7B model on one GPU beats a 70B model you can't run.

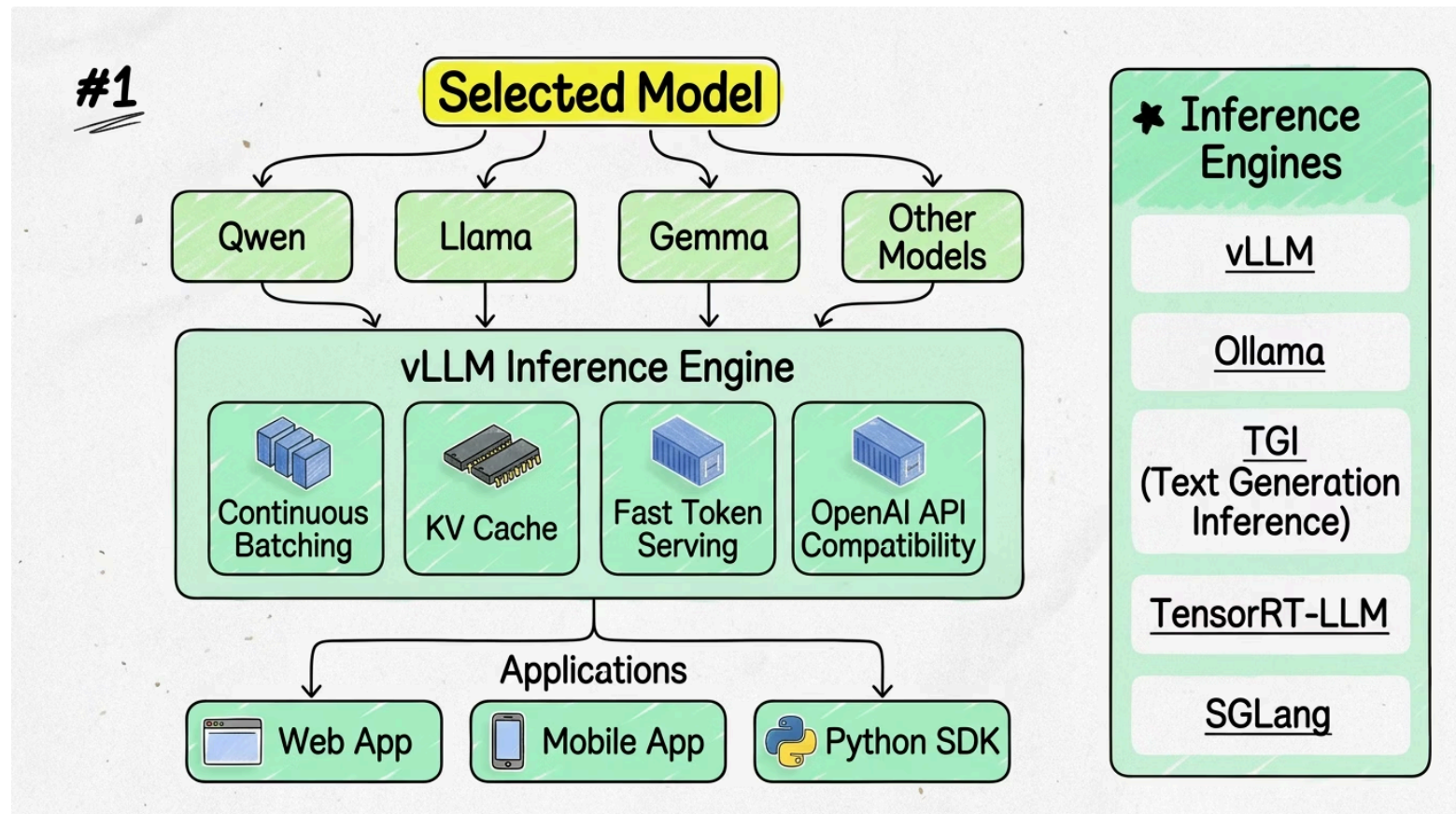


## ⚙️ Step 2

### Choose an Inference Engine

Models need a serving layer before they can handle requests.

vLLM sits between your model and your users — handling batching, caching, and the API.



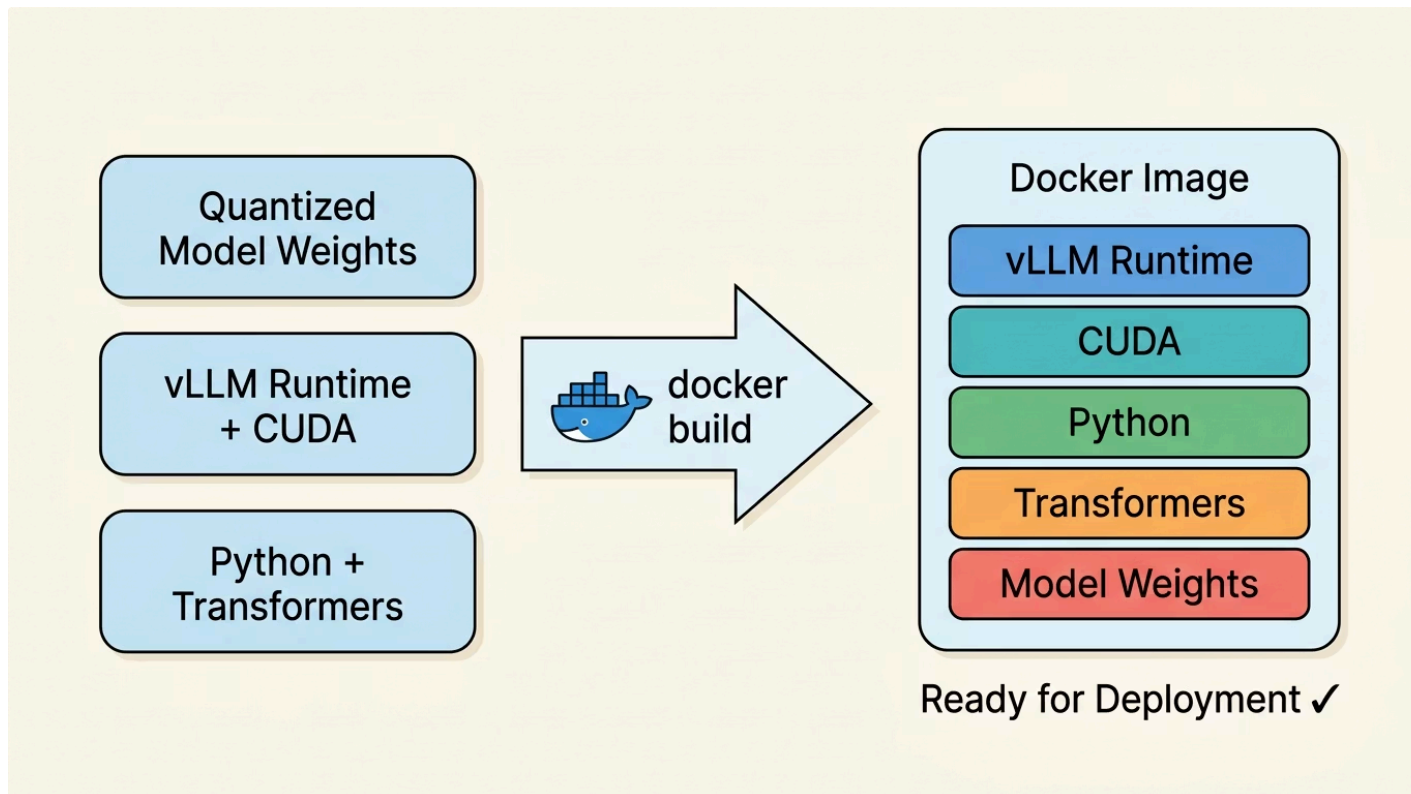


## Step 3

### Package with vLLM

Wrap everything into one portable Docker image.

vLLM + CUDA + model weights = one image. Deploy anywhere.

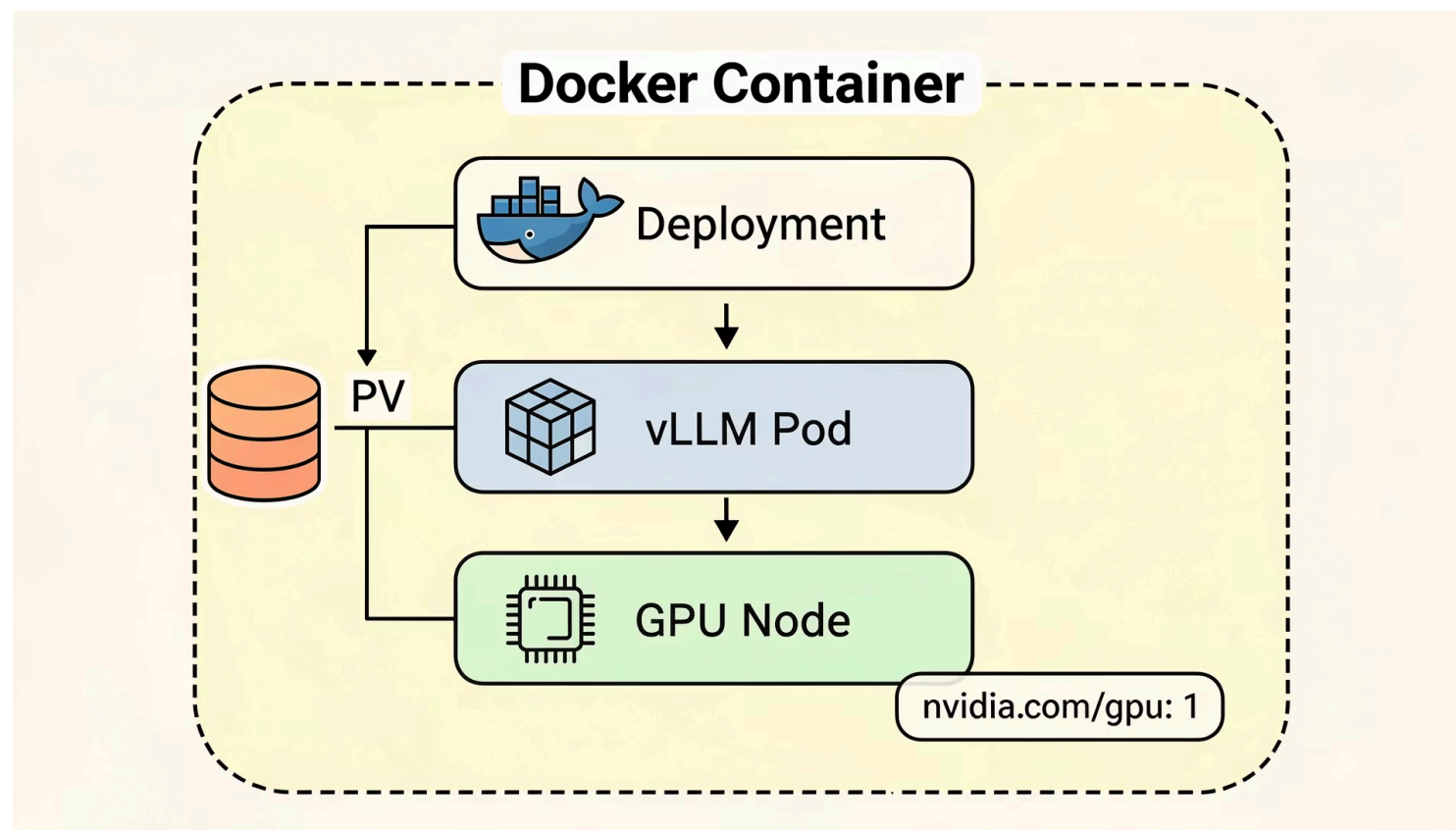


## Step 4

### Docker Container

Run the Docker container directly on a GPU-enabled machine.

One container holds it all — Streamlit, vLLM, model weights, and CUDA runtime.

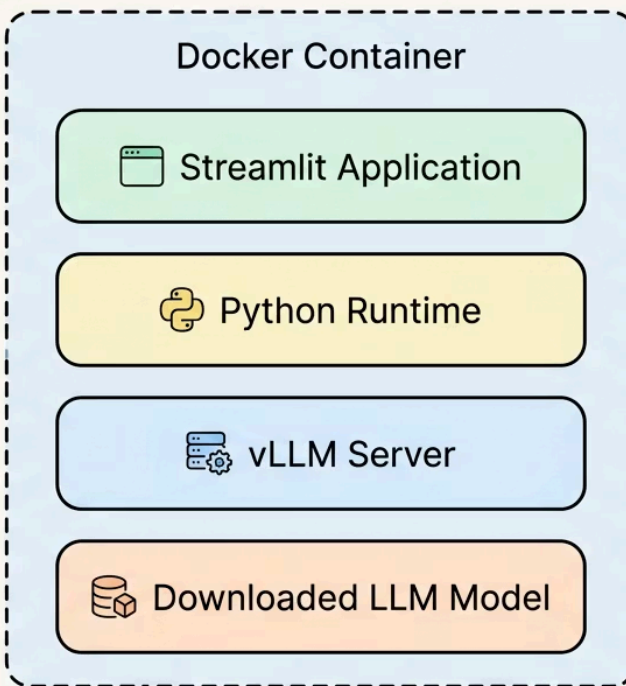


## Step 5

### Single Container Deployment

Package everything into one container — Streamlit, vLLM, and the model together.

One container. One command. Streamlit + vLLM + model weights — all inside.



## Step 6

### Deployment Target & User Access

Run the container on a GPU VM or cloud instance — users connect via browser.

Browser → Streamlit → vLLM → GPU. One VM, one container, full inference pipeline.

